PROBLEM SCOPING

Policy Compliance Checker

Organization Scenario

A university, company, or organization receives thousands of documents such as project reports, research papers, applications, legal documents, and official forms every year. These documents must comply with predefined organizational policies and guidelines before they are accepted.

However,

- Many submitted documents do not comply with organizational policies.
- Important sections are often missing.
- Documents may contain incorrect or conflicting information.
- Manual verification consumes significant time and effort.
- Human reviewers may overlook policy violations.
- Delayed approvals reduce organizational efficiency.

The organization approaches an AI team to develop an intelligent system that can automatically verify whether submitted documents comply with predefined policies and provide instant feedback before final submission.

Solution

Understand the Business Problem

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Problem Statement

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Need of AI

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Identify Stakeholders

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Defining the Task of AI

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Defining Success Metrics

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Defining Constraints

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Assumptions and Risks

↓

Statistical Approach

Step 1 : Understand the Business Problem

Initial Statement

"Many submitted documents do not comply with organizational policies, resulting in delays, rejections, and increased manual verification effort."

The actual problem needs to be analyzed and converted into a measurable AI problem.

Questions for Stakeholders

To Compliance Department

- What is the current document compliance rate?
- Which policy violations occur most frequently?
- Which documents are rejected most often?
- How much time is required for manual verification?

To Policy Experts

- Which policies are mandatory?
- Which violations are considered critical?
- How frequently are policies updated?
- Are there different policies for different document types?

To Users

- Which policy requirements are difficult to understand?
- Which sections are commonly forgotten?
- What type of feedback would help users?
- What challenges do users face while preparing documents?

To Management

- Why is improving compliance important?
- What is the target compliance percentage?
- What budget is available for automation?
- How will the success of the AI system be measured?

Suppose the Organization Provides the Following Reports

- Current compliance rate: **68%**
- Desired compliance rate: **95%**
- Most rejected documents have missing mandatory sections.
- Policy violations increase document approval time.
- Manual verification requires approximately **15–20 minutes** per document.
- Around **15,000** documents are reviewed annually.
- Early detection of policy violations can significantly improve document quality.

Step 2 : Problem Statement

Develop an AI-powered **Policy Compliance Checker** that automatically verifies whether a submitted document follows predefined organizational or academic policies before final submission.

What?

Predict whether a document is compliant or non-compliant.

When?

Before the final document submission.

Why?

To provide early feedback, reduce manual verification, and improve document quality.

Outcome

- Increase policy compliance.
- Reduce document rejection.
- Improve organizational efficiency.

- Reduce manual effort.

Step 3 : Need of AI

Can the problem be solved without AI?

Simple rule-based systems can detect basic issues such as:

- Missing Cover Page
- Missing Signature
- Missing Mandatory Section

Example:

If Signature Missing → Reject Document

However, modern documents contain natural language, different writing styles, and complex policy requirements.

Therefore, manual rules are insufficient.

Why AI is Needed?

Document compliance depends on multiple factors:

1. Mandatory Sections
2. Policy Rules
3. Context of Information
4. Conflicting Statements
5. Document Structure
6. Writing Quality
7. Organization Guidelines
8. Semantic Similarity
9. Missing Information
10. Policy Exceptions

- Policies are complex.
- Multiple rules interact.
- Manual verification is slow.
- AI can understand document meaning using NLP.

<u>WITHOUT AI</u>

Users

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Submit Documents

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Manual Verification

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Long Review Time

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Higher Errors

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Delayed Approval

<u>WITH AI</u>

Users

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Policy Compliance Checker

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Policy Verification

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Compliance Score

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Instant Feedback

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Improved Document Quality

Step 4 : Identify Stakeholders

Organization Management

Improve policy compliance.

Users

Receive instant feedback before submission.

Compliance Team

Reduce manual verification workload.

Policy Experts

Maintain organizational policies.

Review Team

Approve documents efficiently.

IT Department

Deploy and maintain the AI system.

Organization

Improve efficiency and reduce operational cost.

Step 5 : Defining the Task of AI

<u>Machine Learning Formulation</u>

Classification Problem

Output

- Compliant
- Non-Compliant

(Binary Classification)

Risk Scoring

Compliance Score

0% – 100%

Example

- Document A = 98%
- Document B = 74%
- Document C = 36%

Ranking Problem

Which documents require immediate review?

Rank 1 → 18% Compliance

Rank 2 → 30% Compliance

Rank 3 → 42% Compliance

Step 6 : Defining Success Metrics

Business Metrics

- Increase compliance rate from **68%** to **95%**
- Reduce manual verification time.
- Improve document quality.
- Reduce policy violations.

Technical Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Target

Accuracy > 90%

Recall > 85%

Step 7 : Defining Constraints

Operational Constraints

- Limited verification staff.
- Limited implementation budget.

Time Constraints

- Documents must be verified before final submission.

Data Constraints

- Missing policy information.
- Different document formats.
- Poor document quality.
- Incomplete uploaded documents.

Ethical Constraints

- User privacy must be protected.
- AI should assist decisions, not replace humans.
- The system should not be biased.
- Users should receive explainable feedback.

Step 8 : Assumptions and Risks

Assumptions

- Policies are accurate and up to date.
- Historical compliance data is reliable.
- Users upload readable documents.
- Organizations regularly update policies.

Risks

- Missing policy rules.
- Poor-quality scanned documents.
- Frequent policy changes.
- Incorrect AI predictions.
- Data privacy concerns.
- False Positive and False Negative predictions.

Statistical Approach

Historical Documents

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Data Collection

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Text Preprocessing

(Tokenization, Stop-word Removal)

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Feature Extraction

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Document Embedding / NLP Features

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Classification Model

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Compliance Prediction

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Compliance Score & Feedback

Expected Outcome

- Faster document verification.
- Higher compliance rate.
- Reduced manual effort.
- Improved document quality.
- Better organizational efficiency.
- Faster approval process.